

PROJECT NAME: Student Performance Prediction using Decision Tree Classifier

1. Objective

The objective of this project is to develop an end-to-end machine learning model that predicts students' academic performance based on demographic, academic, and behavioral attributes. The project demonstrates the complete machine learning workflow, including data preprocessing, exploratory data analysis (EDA), model building, evaluation, and insight generation.

2. Problem Statement

Educational institutions often face challenges in identifying students who may require additional academic support. Predicting student performance using machine learning techniques can help educators understand the factors influencing academic success and make informed decisions for timely interventions.

The goal of this project is to classify students into Low, Medium, or High performance categories using the Decision Tree Classification algorithm.

3. Dataset Description

The dataset used in this project is the **xAPI-Edu-Data** dataset.

It contains information about students' demographic details, academic behavior, classroom participation, attendance, and parental involvement.

Dataset Information

- Total Records: 480
- Total Features: 17
- Target Variable: Class

The target variable contains three categories:

- L – Low Performance
- M – Medium Performance
- H – High Performance

4. Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

5. Data Preprocessing

The following preprocessing steps were performed before training the model:

- Loaded the dataset using Pandas.
- Explored the dataset using `head()`, `info()`, `describe()`, and `shape()`.
- Checked for missing values.
- Checked and removed duplicate records.
- Converted categorical features into numerical values using Label Encoding.
- Split the dataset into training (80%) and testing (20%) sets.

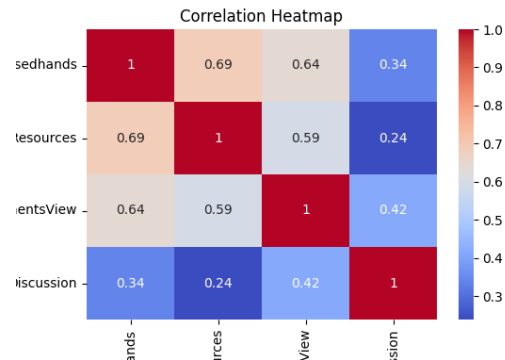
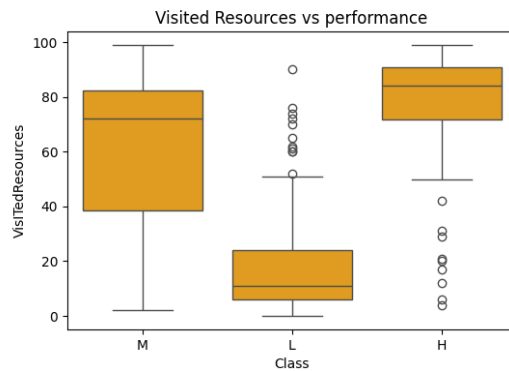
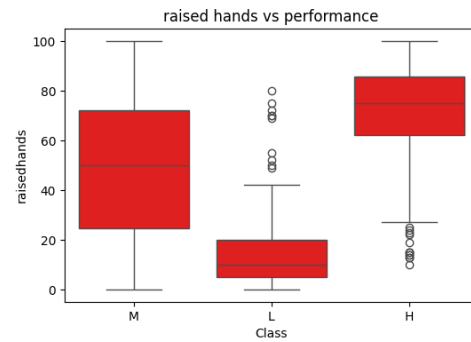
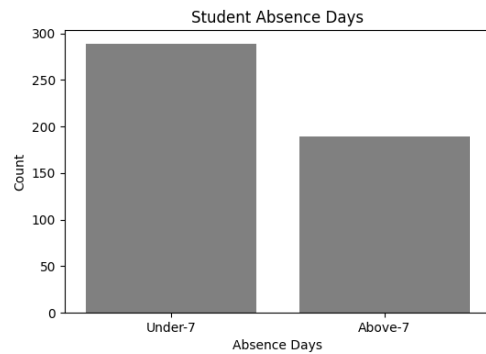
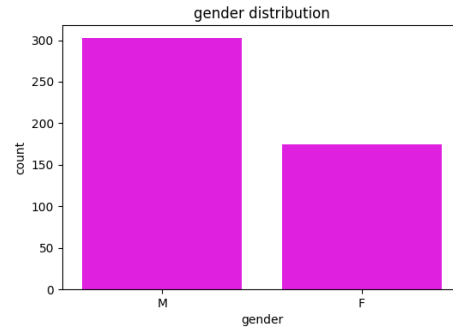
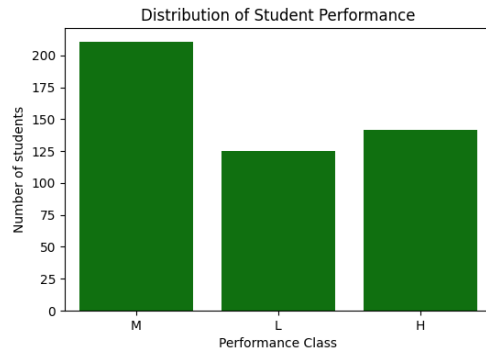
These preprocessing steps ensured that the data was suitable for machine learning algorithms.

6. Exploratory Data Analysis (EDA)

Several visualizations were created to better understand the dataset.

The following graphs were generated:

- Distribution of Student Performance
- Gender Distribution
- Student Absence Days
- Raised Hands vs Performance
- Visited Resources vs Performance
- Correlation Heatmap



Observations

- Student performance was distributed across Low, Medium, and High categories.
- Students with fewer absence days generally performed better.
- Higher classroom participation was associated with improved academic performance.
- Students who frequently visited learning resources tended to achieve better results.
- Positive relationships were observed among several engagement-related features.

7. Model Building

A Decision Tree Classifier was selected for this project because it is simple, interpretable, and effective for classification problems.

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

The Decision Tree model was trained using the training data and evaluated on the testing data.

8. Model Evaluation

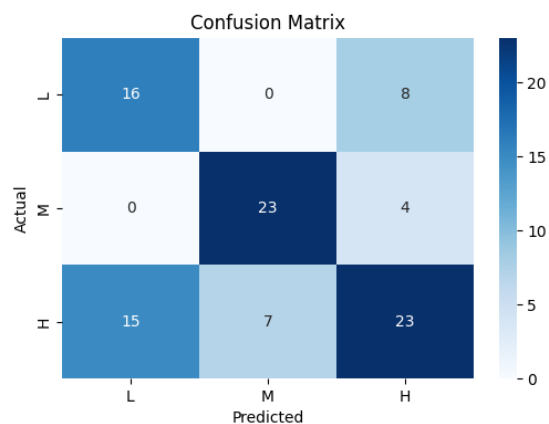
The model performance was evaluated using the following metrics:

- Accuracy Score
- Classification Report
- Confusion Matrix
- Feature Importance

Model Accuracy

Accuracy = **64.58%**

The model correctly classified approximately 64.58% of the students in the testing dataset.

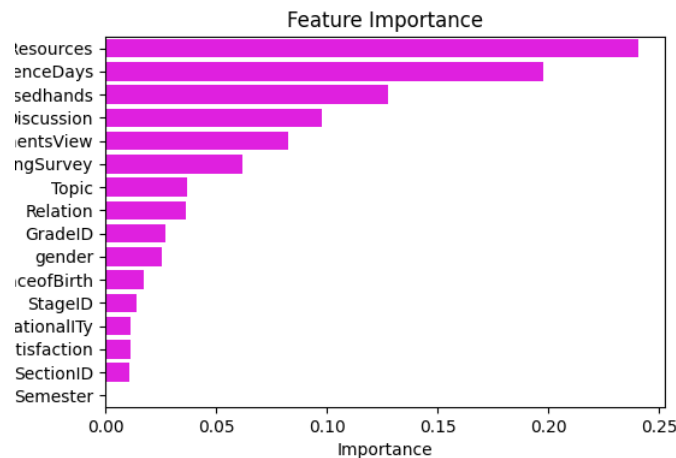


9. Feature Importance

Feature importance analysis identified the variables that contributed most to predicting student performance.

Top five important features:

Feature	Importance
VisITedResources	0.2404
StudentAbsenceDays	0.1979
raisedhands	0.1278
Discussion	0.0979
AnnouncementsView	0.0826



10. Results

The Decision Tree model successfully classified student performance into Low, Medium, and High categories.

The analysis showed that behavioral factors had a stronger influence on academic performance than demographic characteristics.

The model demonstrated that attendance, participation, and engagement significantly contribute to predicting student success.

11. Key Insights

- Students who visited learning resources more frequently generally achieved better academic performance.
- Regular attendance positively influenced student success.
- Active classroom participation, measured through raised hands and discussions, contributed to improved performance.
- Students who regularly viewed course announcements showed greater engagement in learning.
- Behavioral and engagement-related features were more influential than demographic attributes such as gender or nationality.

12. Conclusion

This project successfully demonstrated the complete machine learning workflow for predicting student academic performance.

The Decision Tree Classifier achieved an accuracy of **64.58%** and identified meaningful patterns in the dataset. The results indicate that classroom engagement, attendance, and effective use of learning resources play an important role in students' academic success.

This project highlights how machine learning can assist educational institutions in identifying students who may require additional academic support and improving educational outcomes.